

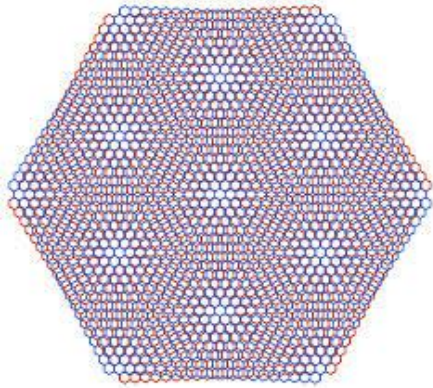
First-Principles Calculations of Moiré Excitons in Twisted Two-Dimensional Materials

Hongli Guo^{1,2}, Xu Zhang², Gang Lu²

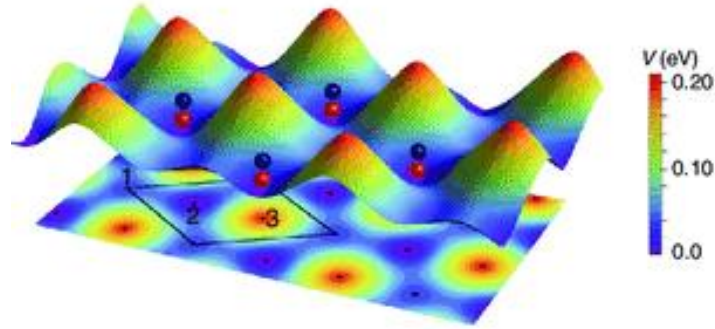
¹School of Physics, Zhejiang University

²California State University Northridge

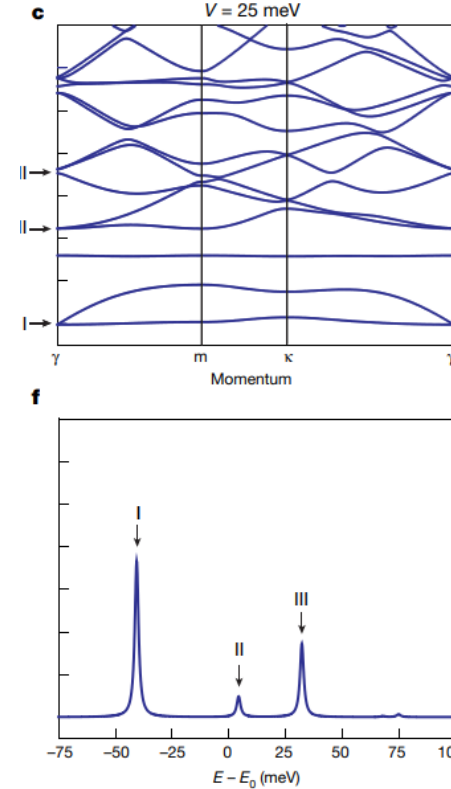
Moiré Excitons in twisted 2D materials



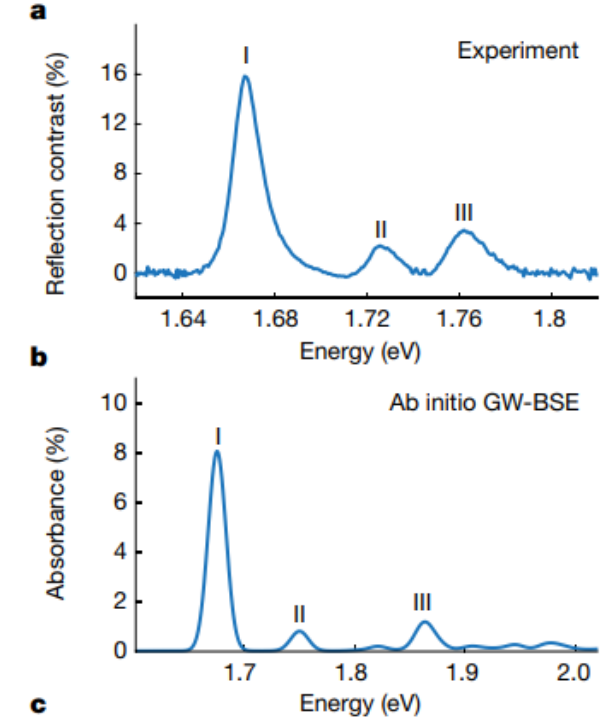
moiré lattice



moiré potential and moiré excitons



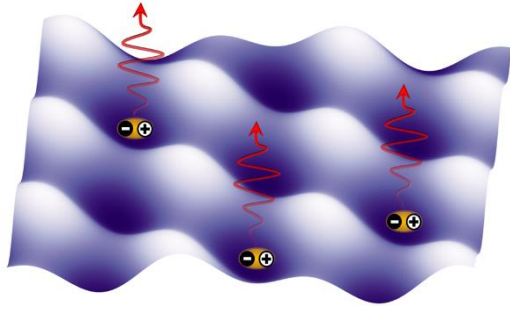
Jin, C., Nature 2019



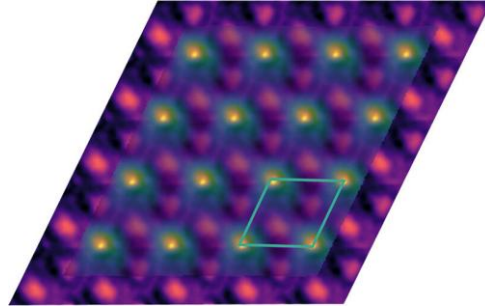
Mit H. Maik, Nature 2022

Understanding these unique excitonic properties in twisted 2D materials opens new avenues toward advanced optoelectronics and quantum technologies.

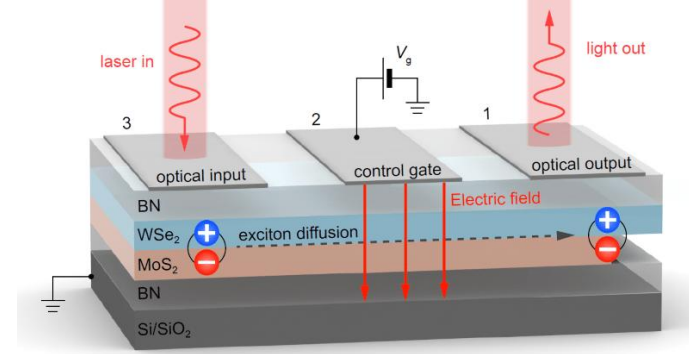
Moiré Excitons in twisted 2D materials



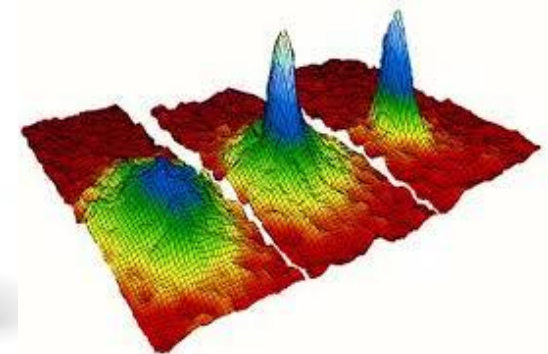
Single photon emission



Quantum Computing



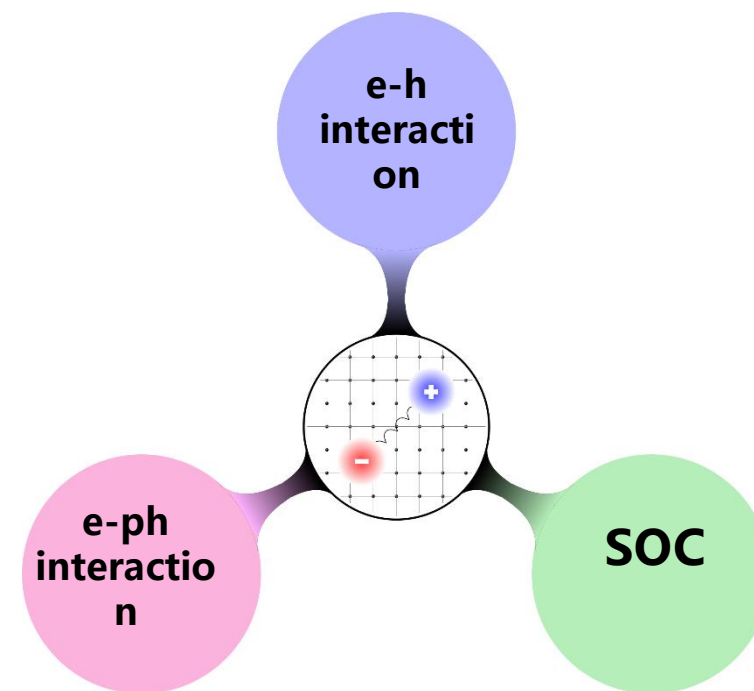
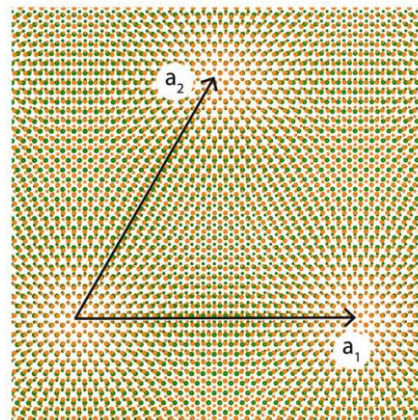
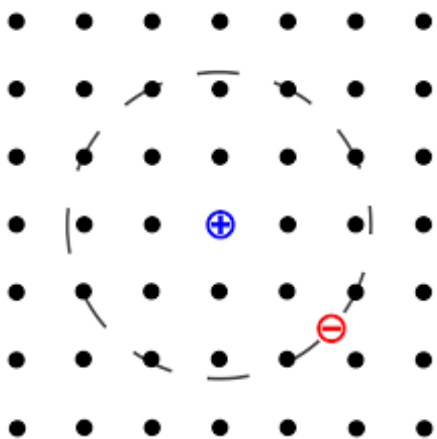
Exciton devices



BEC

Can we achieve accurate, predictive understanding of moiré excitons through advanced first-principles methods?

Challenge in First-principles calculations of Moiré Excitons



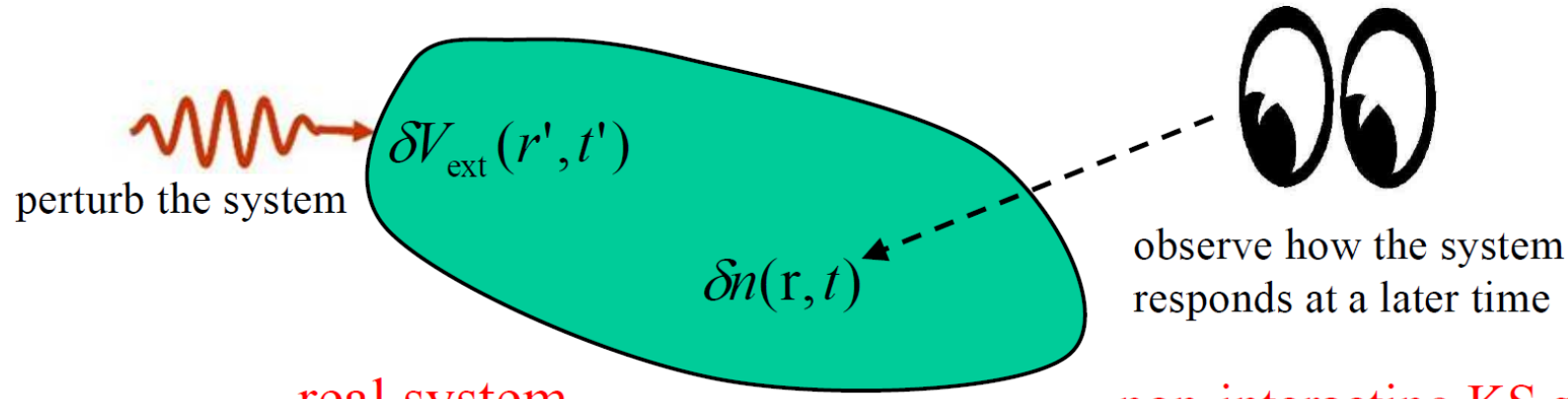
➤ Electron-hole interaction

➤ Large scale

➤ Different couplings

- ❖ GW-BSE: Accurate、**Huge computational demanding**
- ❖ Conventional LR-TDDFT: can't accurately calculate binding energy、**underestimate band gap**

Linear Response Time-Dependent Density Functional Theory (LR-TDDFT)



real system

non-interacting KS system

$$\delta n(r; \omega) = \int dr' \chi[n_0](r, r'; \omega) \delta V_{\text{ext}}(r'; \omega) = \int dr' \chi_{\text{KS}}[n_0](r, r'; \omega) \delta V_{\text{KS}}(r'; \omega)$$

$$\chi^{-1}(\omega) = \chi_{\text{KS}}^{-1}(\omega) - K_{\text{HXC}}(\omega)$$

Poles: true many-body excitations

Poles: single-particle KS excitations

Hartree and XC kernel: $K_{\text{HXC}}(\omega) = 1/|r - r'| + f_{\text{xc}}[n_0](r, r'; \omega)$

Exact formulation, but in practice approximations needed

Adiabatic approximation: no time or ω -dependence in the kernel term

$$f_{\text{xc}}[n_0](r, r'; \omega)$$

LDA, GGA, hybrids

LR-TDDFT+screened range separated hybrid functional

Casida equation:

$$\begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{B}^* & \mathbf{A}^* \end{pmatrix} \begin{pmatrix} \mathbf{X}_I \\ \mathbf{Y}_I \end{pmatrix} = \omega_I \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} \mathbf{X}_I \\ \mathbf{Y}_I \end{pmatrix}$$

$$\Phi_I \approx \sum_{ij\sigma} \frac{X_{I,ij\sigma} + Y_{I,ij\sigma}}{\sqrt{\omega_I}} \hat{a}_{j\sigma}^+ \hat{a}_{i\sigma} \Phi_0$$

(1) Using screened and range-separated hybrid (SRSH) XC functionals

Coulomb repulsion operator:

$$\frac{1}{r} = \frac{\alpha + \beta \text{erf}(\mu r)}{r} + \frac{1 - \alpha - \beta \text{erf}(\mu r)}{r}$$

$\alpha + \beta = 1/\epsilon_0$; α and μ can be tuned to reproduce fundamental band gap (from either experiments or GW)

SRSH XC potential:

$$V_{\text{xc}}^{\text{RSH}} = \alpha V_{\text{Fx}} + \beta V_{\text{Fx}}^{\text{LR}} + (1 - \alpha) V_{\text{KSx}} - \beta V_{\text{KSx}}^{\text{LR}} + V_{\text{KSc}}$$

(2) Using first-order perturbation to solve the SRSH Hamiltonian

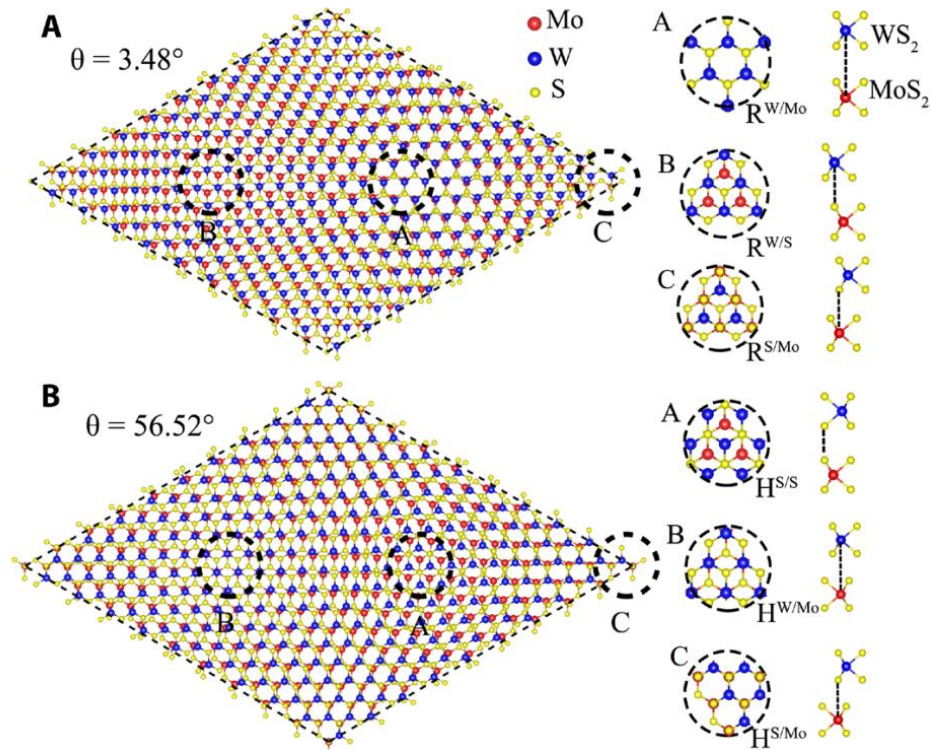
$$\hat{H}^{\text{RSH}} = \hat{H}^{\text{GGA}} + \Delta \hat{H} \quad \Delta \hat{H} = \hat{V}_{\text{xc}}^{\text{RSH}} - \hat{V}_{\text{xc}}^{\text{GGA}}$$

Quasiparticle energy levels:

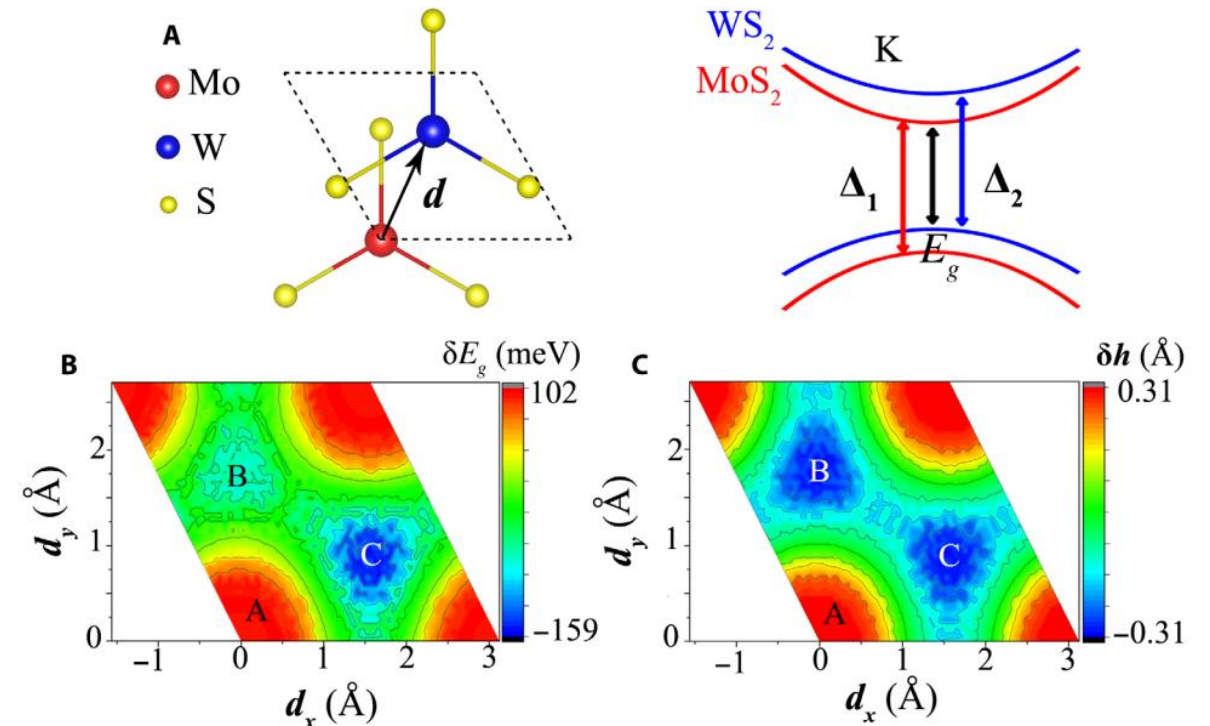
$$\epsilon_i^{\text{RSH}} = \epsilon_i^{\text{GGA}} + \langle \varphi_i^{\text{GGA}} | \Delta \hat{H} | \varphi_i^{\text{GGA}} \rangle$$

Deep moiré potential in twisted WS₂/MoS₂

1626 atoms

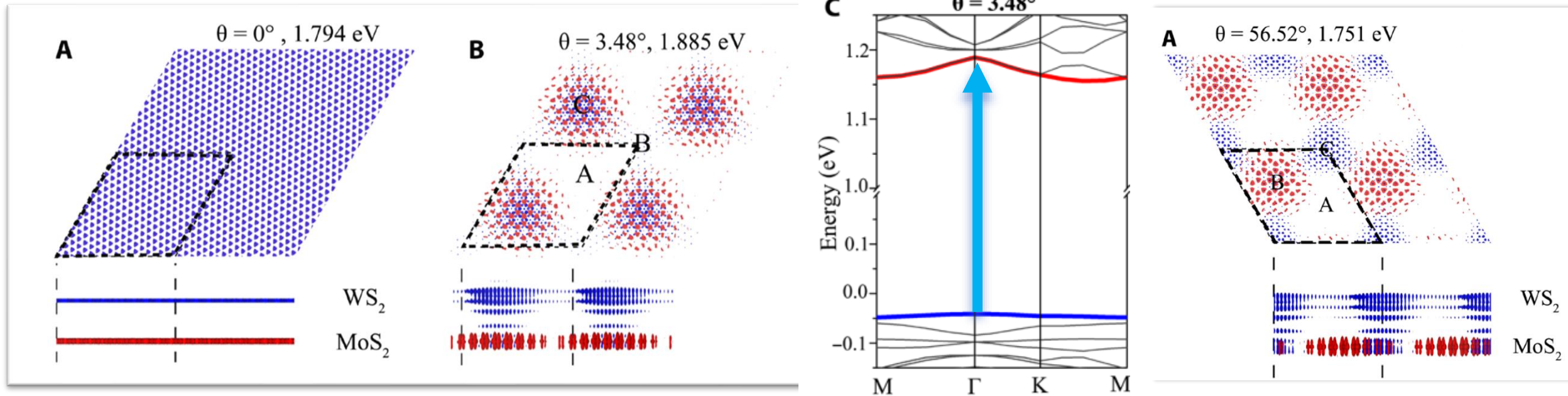


6 stackings in moiré patterns



Moiré potential mapping

Shedding light on moiré excitons: A first-principles perspective



Charge-transfer moiré exciton

$$\langle \Phi_I | \hat{\rho}(\mathbf{r}) | \Phi_I \rangle = \rho_0(\mathbf{r}) + \sum_{i,j,j'} z_{I,ij}^* z_{I,ij'} \phi_j^*(\mathbf{r}) \phi_{j'}(\mathbf{r}) - \sum_{ii',j} z_{I,ij}^* z_{I,i'j} \phi_{i'}^*(\mathbf{r}) \phi_i(\mathbf{r}),$$

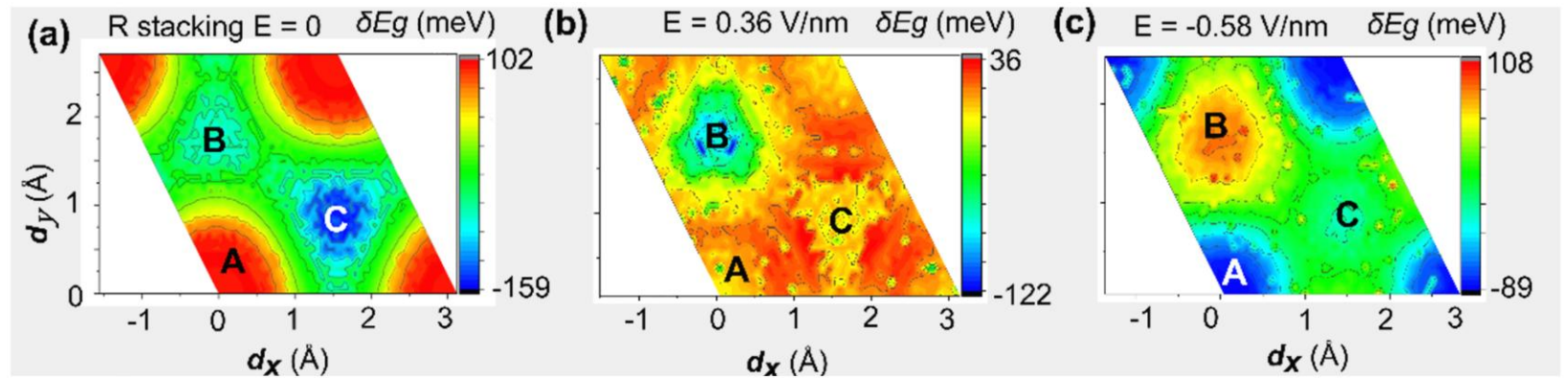
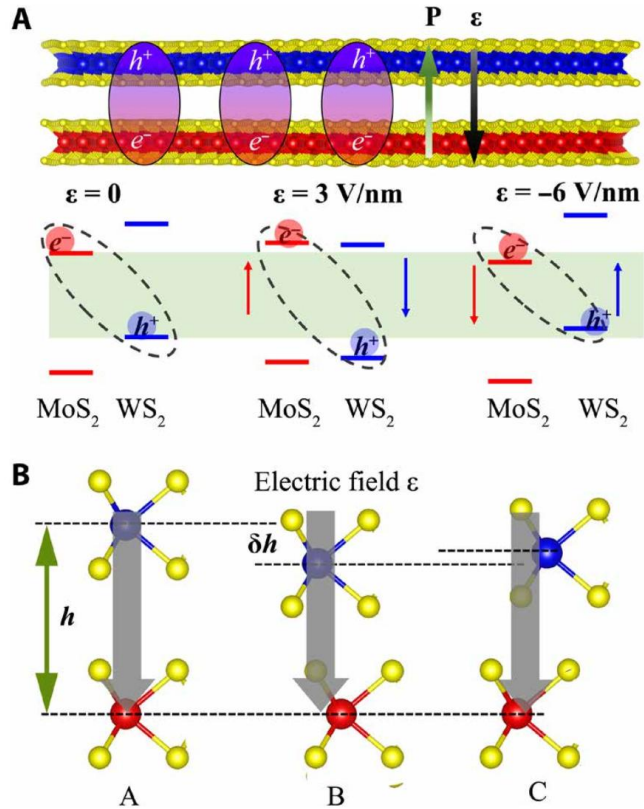
Charge densities of moiré excitons for heterostructures with twist angle $\theta = 3.48^\circ$ (left) and $\theta = 56.52^\circ$ (right)

Red: electron

Blue: hole

Binding energy: 207 meV

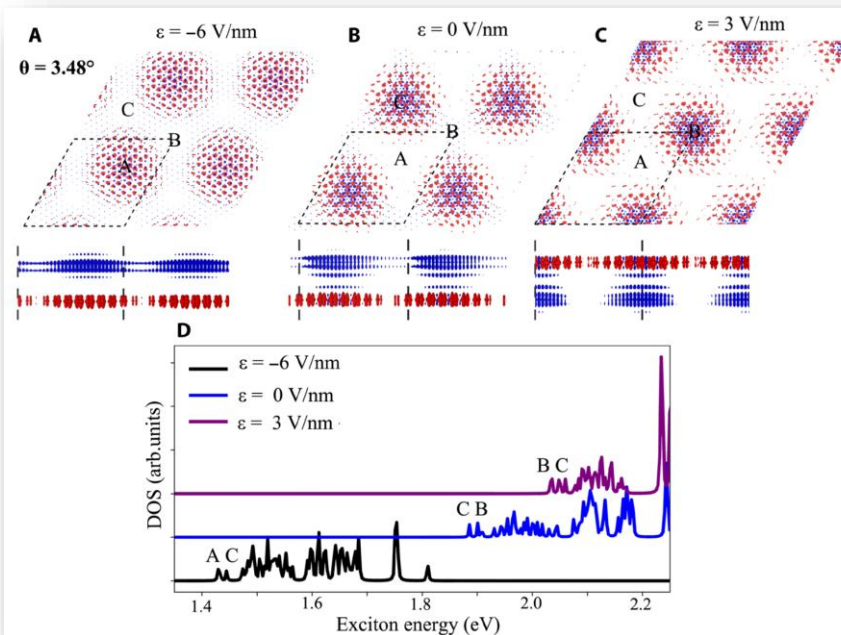
Electrical field tuning of moiré excitons



260 meV

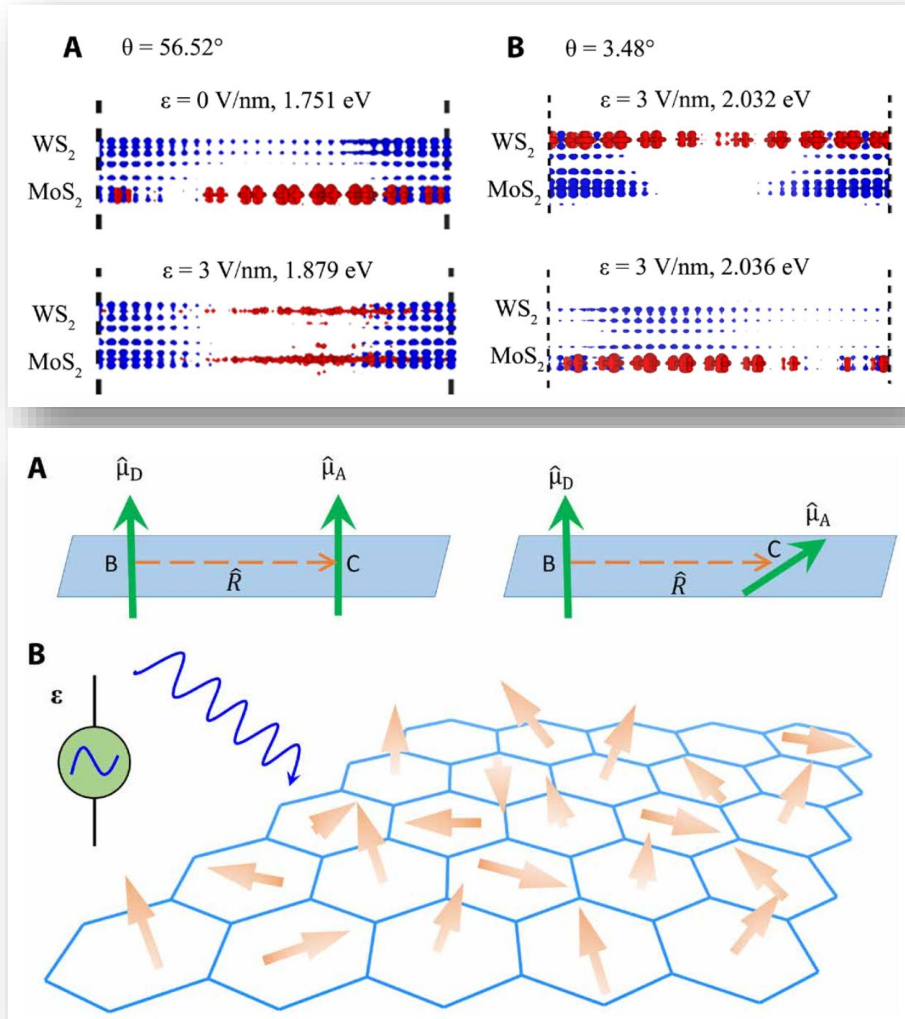
168 meV

197 meV



Electrical field tuning of moiré excitons: position, energy and polarity

Tuning of the hybridization of moiré excitons and exciton lifetime



Electrical field tuning of the hybridization of moiré excitons

The hybridization changes the dipole moment of the exciton,

Leading to the change of the exciton lifetime and diffusion length

Based on Förster resonance energy transfer theory.

orientation factor

$$\kappa = \hat{\mu}_D \cdot \hat{\mu}_A - 3(\hat{\mu}_D \cdot \hat{R})(\hat{\mu}_A \cdot \hat{R})$$

exciton transition rate

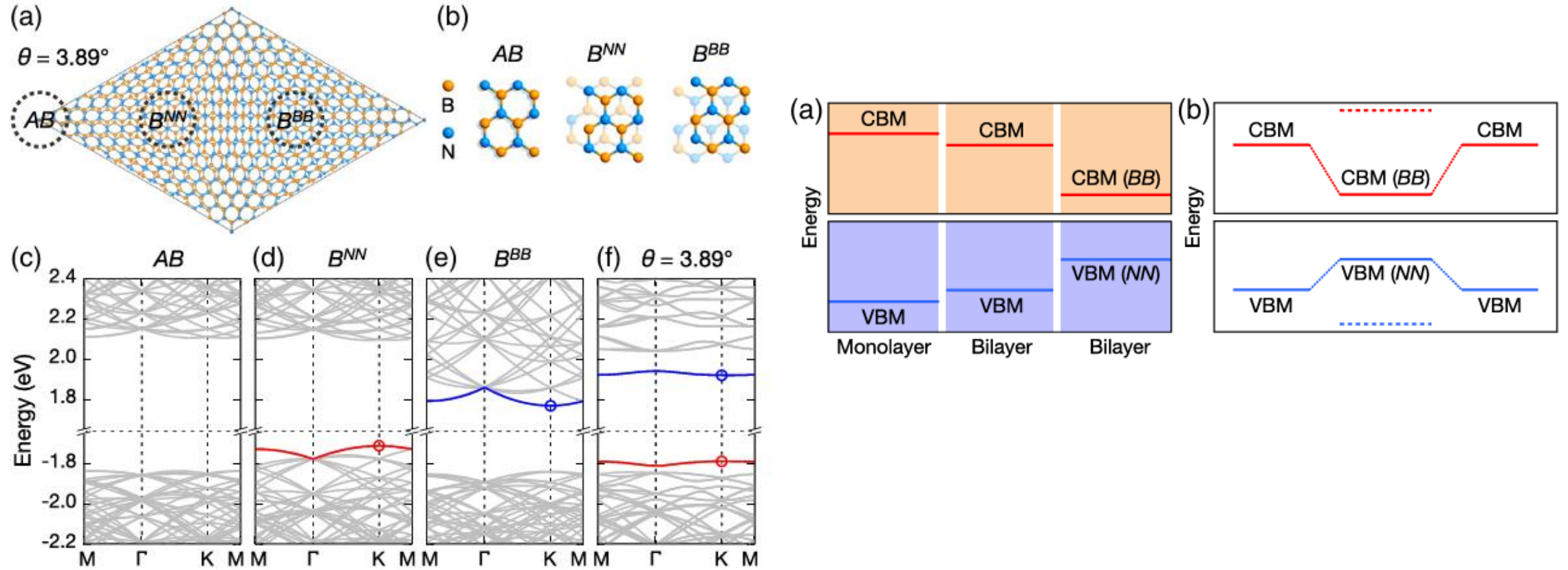
$$k_{IJ} = \frac{1}{\hbar^2} \frac{\kappa^2}{R^6} \mu_I^2 \mu_J^2 \delta(\Delta\omega)$$

spontaneous emission rate

$$k_{I0} = \frac{4n(\omega_i - \omega_j)^3 |\mu_I|^2}{3c^3}$$

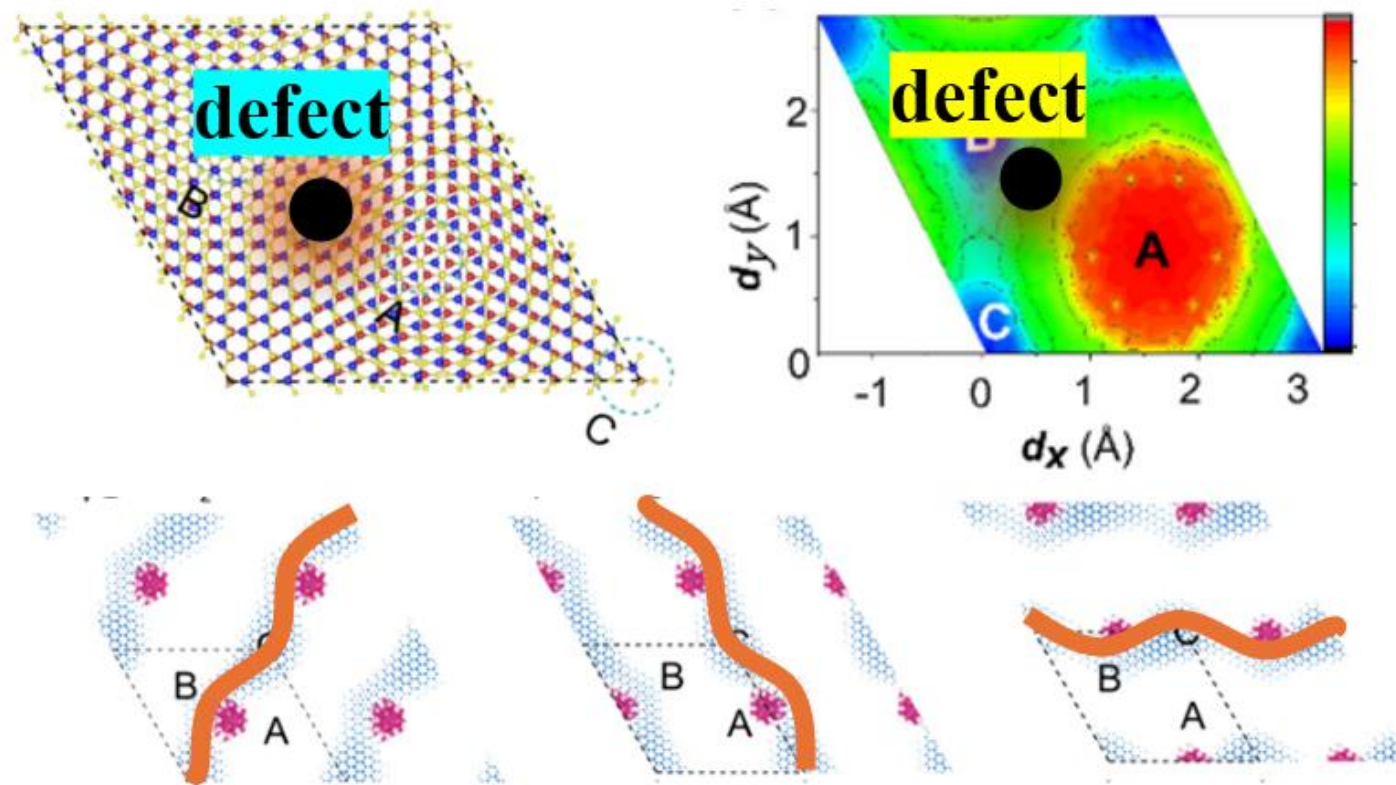
Lifetime $\sim 1\text{ns}$

defectlike moiré flat band



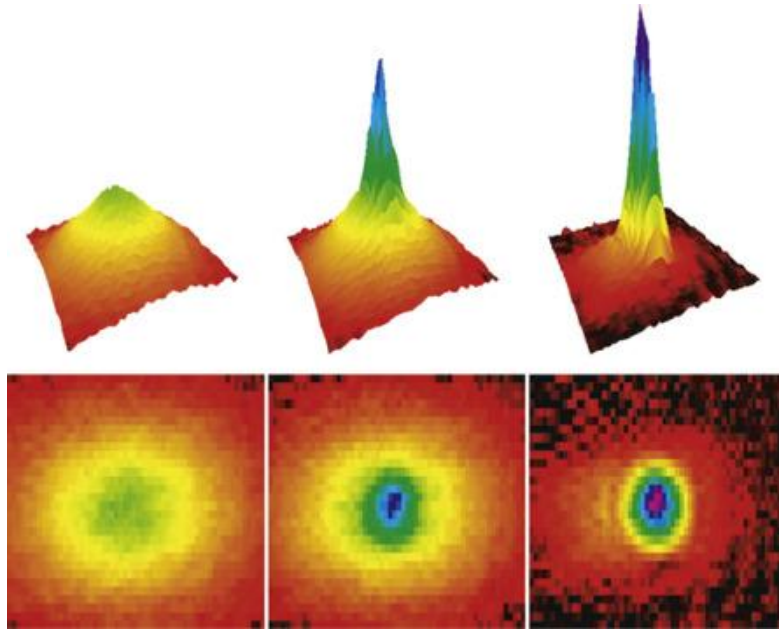
The interplay between defect and moiré potential ?

Moiré excitons in defective van derWaals heterostructures

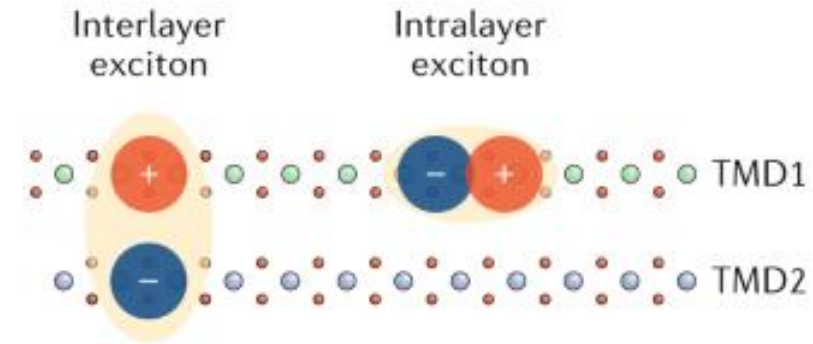


The interplay between defect and moiré potential tunes the moiré exciton
Cooperation and Competition
Quasi 1D excitons

Excitonic Bose-Einstein Condensation (BEC)



Bose-Einstein condensation (BEC) is an exotic quantum state of matter where bosons, such as atoms, can spontaneously condense into a single macroscopic quantum state (T_c for first atomic BEC is nK-uK).



Tartakovskii, A. Nat. Rev. Phys. (2020).

Compared to atoms, excitons have much smaller masses, thus are expected to condense at much higher temperatures

$$T_c = \frac{2\pi\hbar}{k_B m} \left(\frac{n}{2.612} \right)^{2/3}$$

Excitonic Bose-Einstein Condensation (BEC)



ARTICLE

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High-temperature superfluidity with indirect excitons in van der Waals heterostructures

M.M. Fogler¹, L.V. Butov¹ & K.S. Novoselov²

LETTER

<https://doi.org/10.1038/s41586-019-1591-7>

Evidence of high-temperature exciton condensation in two-dimensional atomic double layers

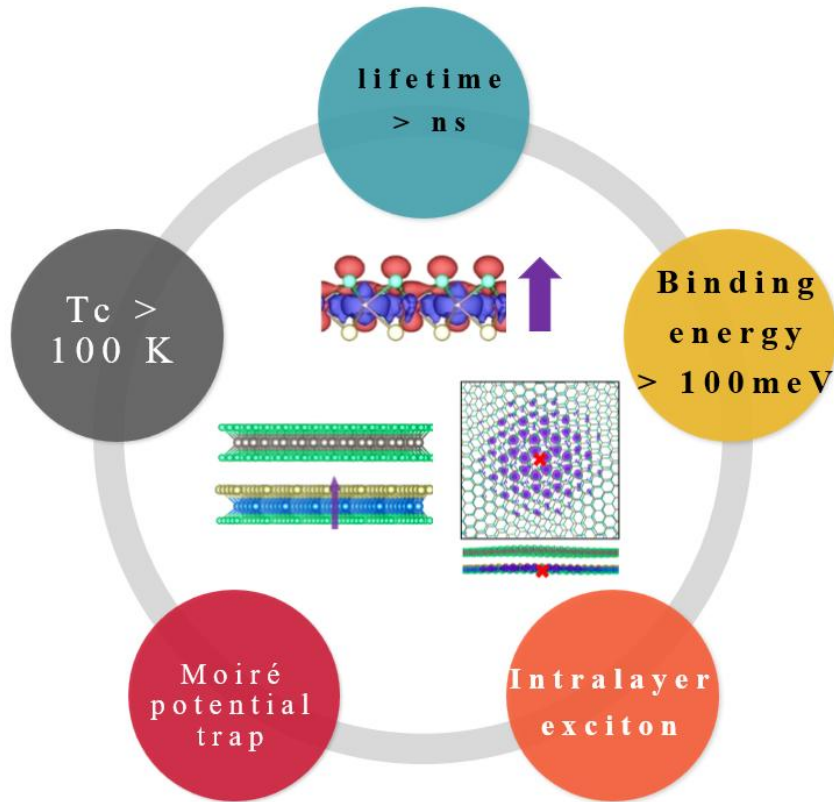
Zefang Wang¹, Daniel A. Rhodes², Kenji Watanabe³, Takashi Taniguchi³, James C. Hone², Jie Shan^{1,4,5*} & Kin Fai Mak^{1,4,5*}



It was predicted that **interlayer** excitons in TMD bilayers could form BEC at ~ 100 K and recently experimental evidence for high temperature (~ 100 K) exciton condensation in MoSe₂/WSe₂ heterobilayers has been reported. In both cases, the BEC is formed by **interlayer** excitons whose electrons and holes are separated by hexagonal boron nitride (hBN) spacer layers.

Whether **moiré excitons** in TMD bilayers (i.e., without a spacer layer) could also condense at high temperatures? Whether **intralayer moiré excitons** with electrons and holes at the same TMD layer could realize high temperature BEC?

Excitonic Bose-Einstein Condensation (BEC)



(moiré potentials could trap excitons)

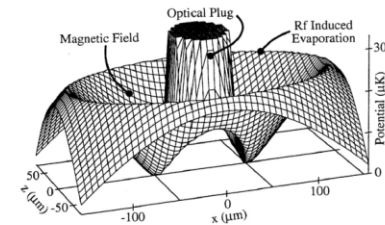
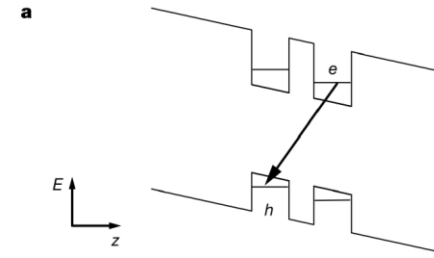
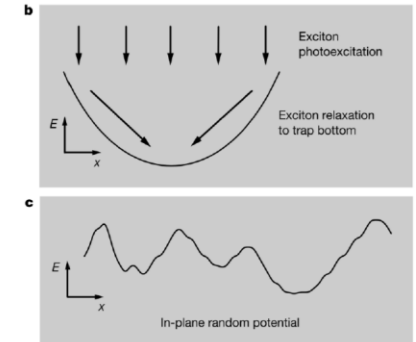


FIG. 1. Adiabatic potential due to the magnetic quadrupole field, the optical plug, and the rf. This cut of the three-dimensional potential is orthogonal to the propagation direction (y) of the blue-detuned laser. The symmetry axis of the quadrupole field is the z axis.



Butov et al. Nature (2002)

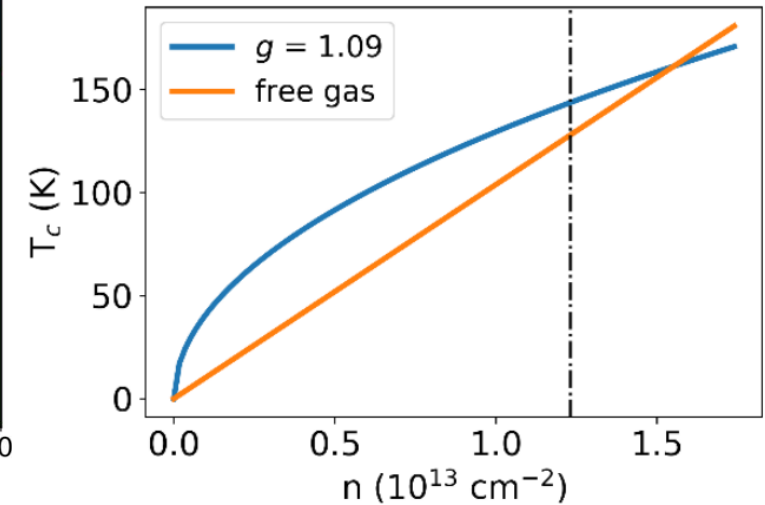
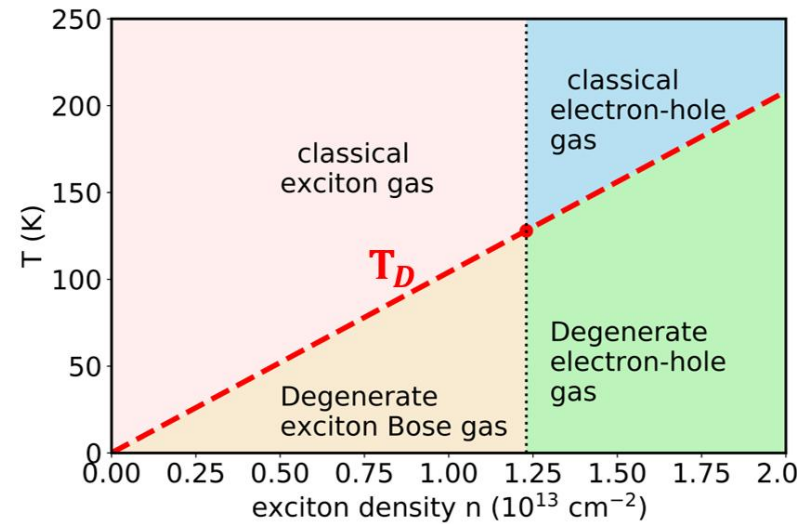
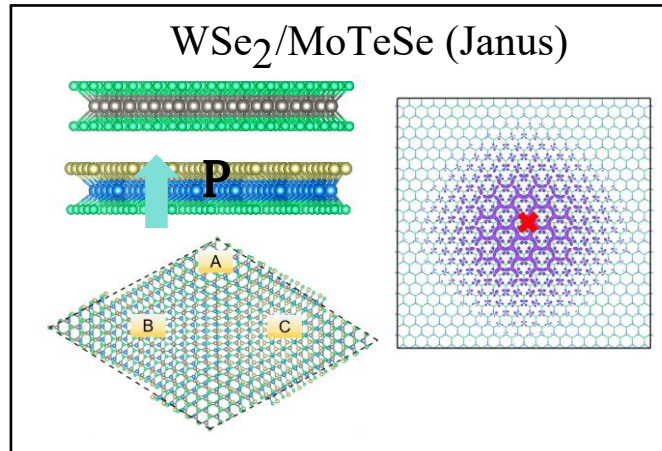


random fluctuation of quantum well thickness and alloy disorder **difficult to control**

- long lifetime
- large binding energy
- potential trap
- dipole interaction

$$T_c = \frac{2\pi\hbar}{k_B m} \left(\frac{n}{2.612} \right)^{2/3}$$

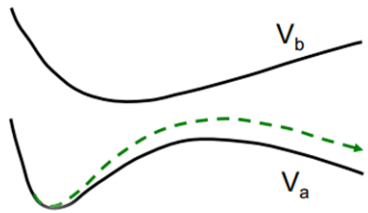
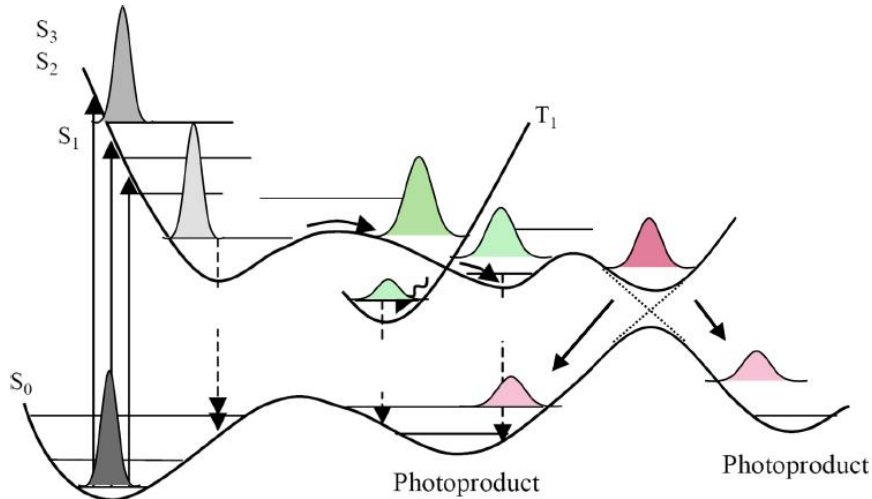
Tuning moiré excitons in Janus heterobilayers for high-temperature BEC



- Exciton lifetime $> \text{ns}$
- Exciton binding energy $> 100 \text{ meV}$
- Intralayer exciton without spacer layer
- Moiré potential trap
- Dipole dipole interaction
- **$T_c > 100 \text{ K}$**

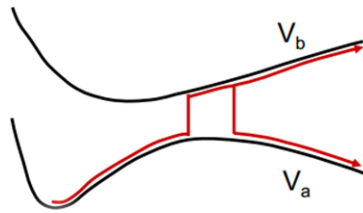
Non-adiabatic Molecular Dynamics (NAMD)

$$\Psi(\mathbf{r}, \mathbf{R}, t) = \Omega_j(\mathbf{R}, t) \Phi_j(\mathbf{r}; \mathbf{R}); \quad \hat{\mathcal{H}}_{el}(\mathbf{r}; \mathbf{R}) \Phi_j(\mathbf{r}; \mathbf{R}) = E_j(\mathbf{R}) \Phi_j(\mathbf{r}; \mathbf{R})$$



Ehrenfest Dynamics

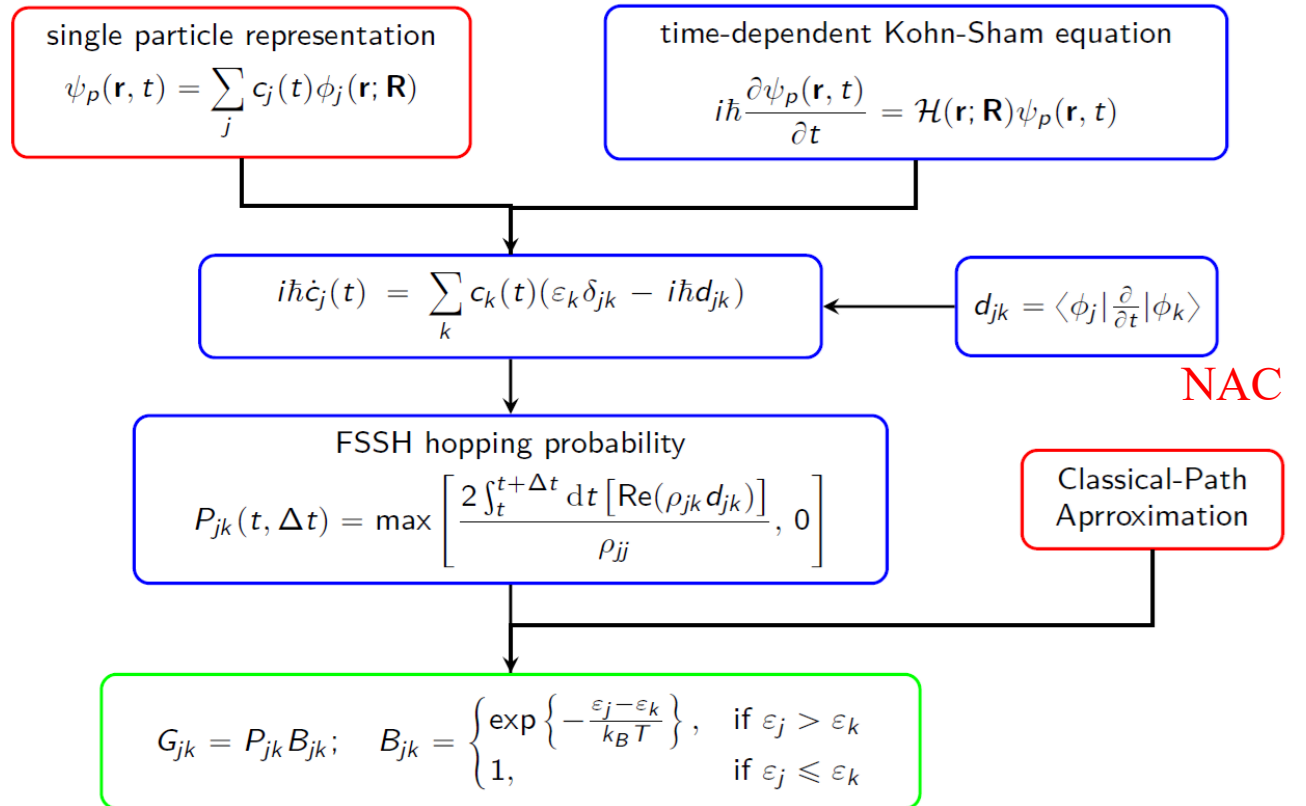
$$M_\alpha \ddot{\mathbf{R}}_\alpha = -\nabla_\alpha \langle \hat{\mathcal{H}}_{el}(\mathbf{r}, \mathbf{R}) \rangle$$



Surface Hopping

$$M_\alpha \ddot{\mathbf{R}}_\alpha = -\nabla_\alpha E_k^{el}(\mathbf{R})$$

Surface hopping scheme



Exciton dynamics: LR-TDDFT + NAMD

Excited state I

$$\Phi_I \approx \sum_{ij\sigma} \frac{X_{I,ij\sigma} + Y_{I,ij\sigma}}{\sqrt{\omega_I}} \hat{a}_{j\sigma}^\dagger \hat{a}_{i\sigma} \Phi_0 = \sum_{ij\sigma} Z_{I,ij} \hat{a}_{j\sigma}^\dagger \hat{a}_{i\sigma} \Phi_0 \quad z_{I,ij} = (X_{I,ij} + Y_{I,ij})/\sqrt{\omega_I}$$

During a molecular dynamics trajectory $\Psi(t) = \sum_{I=0}^{\infty} C_I(t) \Phi_I[R(t)]$

$$\frac{\partial}{\partial t} C_J(t) = - \sum_K C_K(t) \left(\frac{i}{\hbar} \omega_K \delta_{JK} + D_{JK} \right)$$

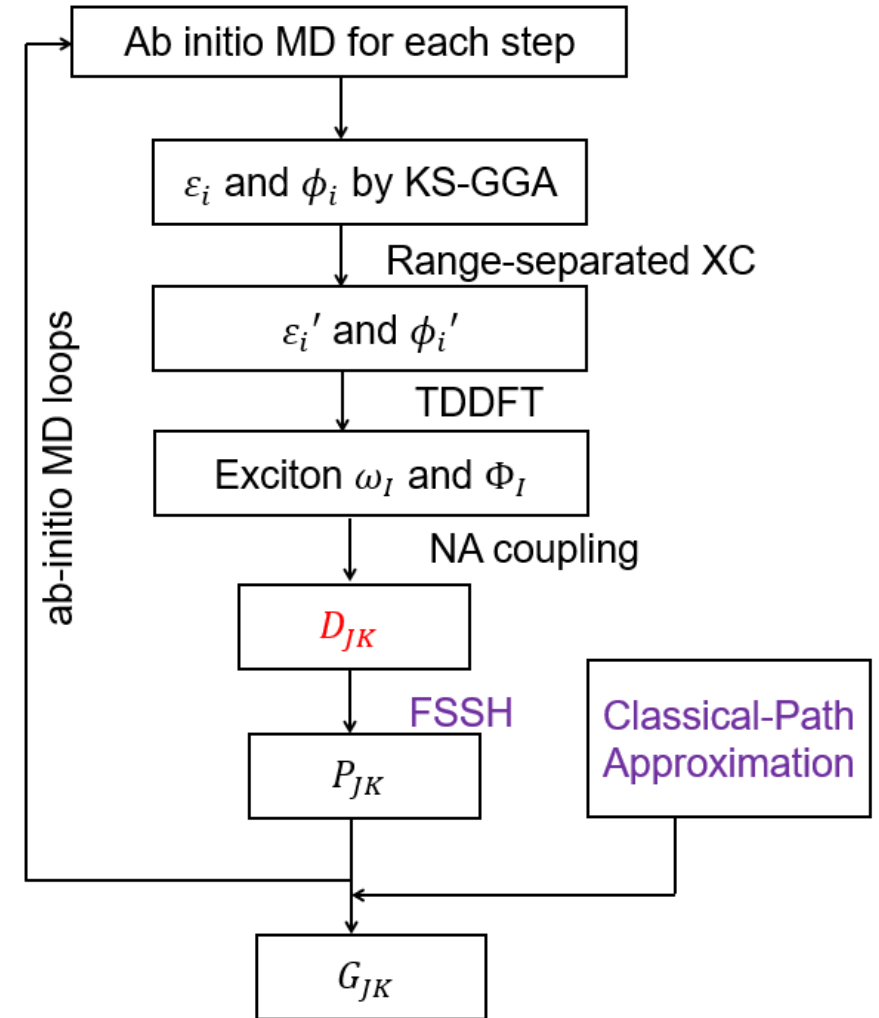
Nonadiabatic coupling (NAC) between two states J and K

$$D_{JK} \equiv \langle \Phi_J | \nabla_R | \Phi_K \rangle \cdot \frac{dR}{dt} = \langle \Phi_J | \frac{\partial}{\partial t} | \Phi_K \rangle$$

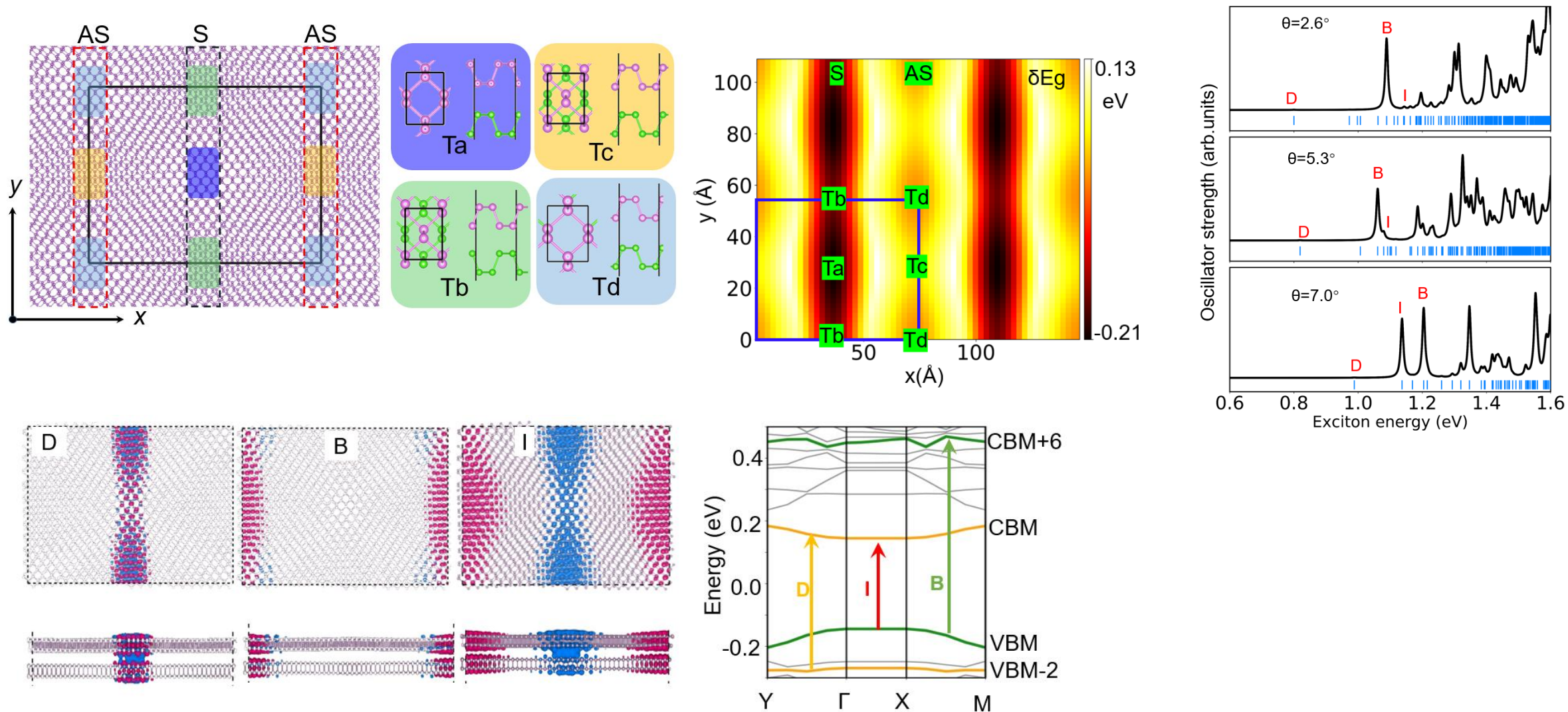
$$D_{JK} = \sum_{ii',jj'} Z_{J,ij}^* Z_{K,i'j'} \langle \Phi_{ij} | \frac{\partial}{\partial t} | \Phi_{i'j'} \rangle = \sum_{i,j \neq j'} Z_{J,ij}^* Z_{K,i'j'} d_{jj'} - \sum_{i \neq i',j} Z_{J,ij}^* Z_{K,i'j'} d_{ii'}$$

$\Phi_{ij} = \hat{a}_j^\dagger \hat{a}_i \Phi_0$ is a single SD of the KS orbitals

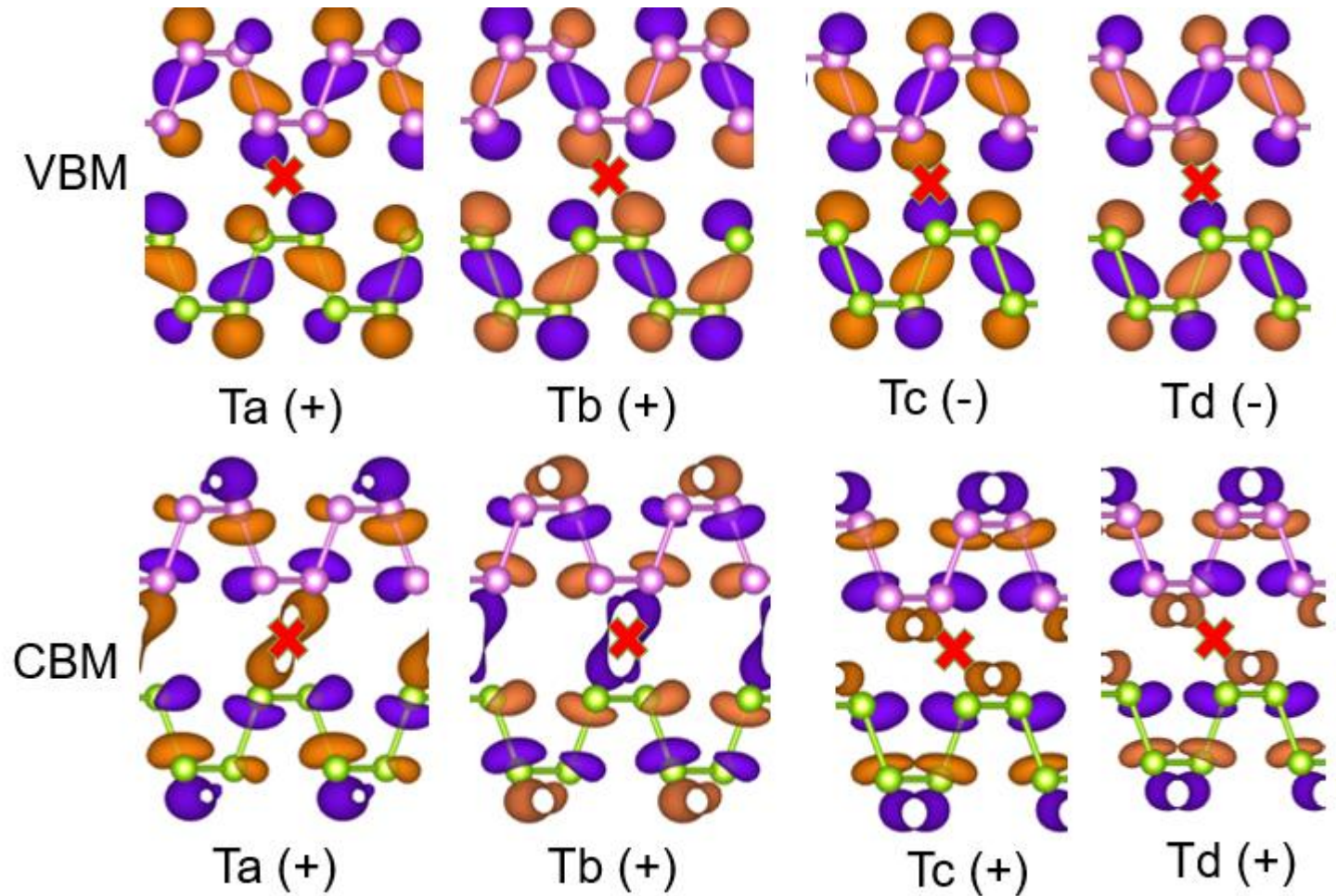
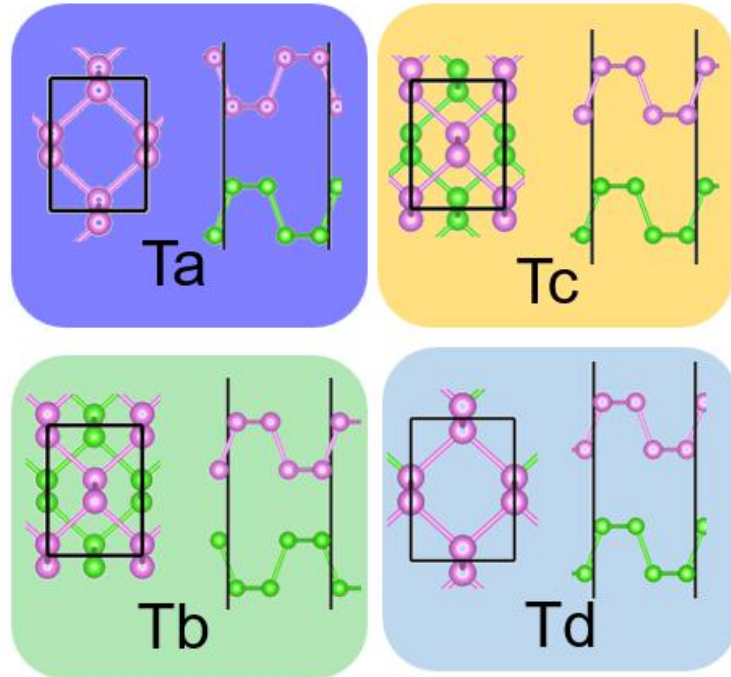
$d_{jk} = \langle \Phi_j | \frac{\partial}{\partial t} | \Phi_k \rangle$ is the nonadiabatic coupling (NAC) between the KS orbitals.



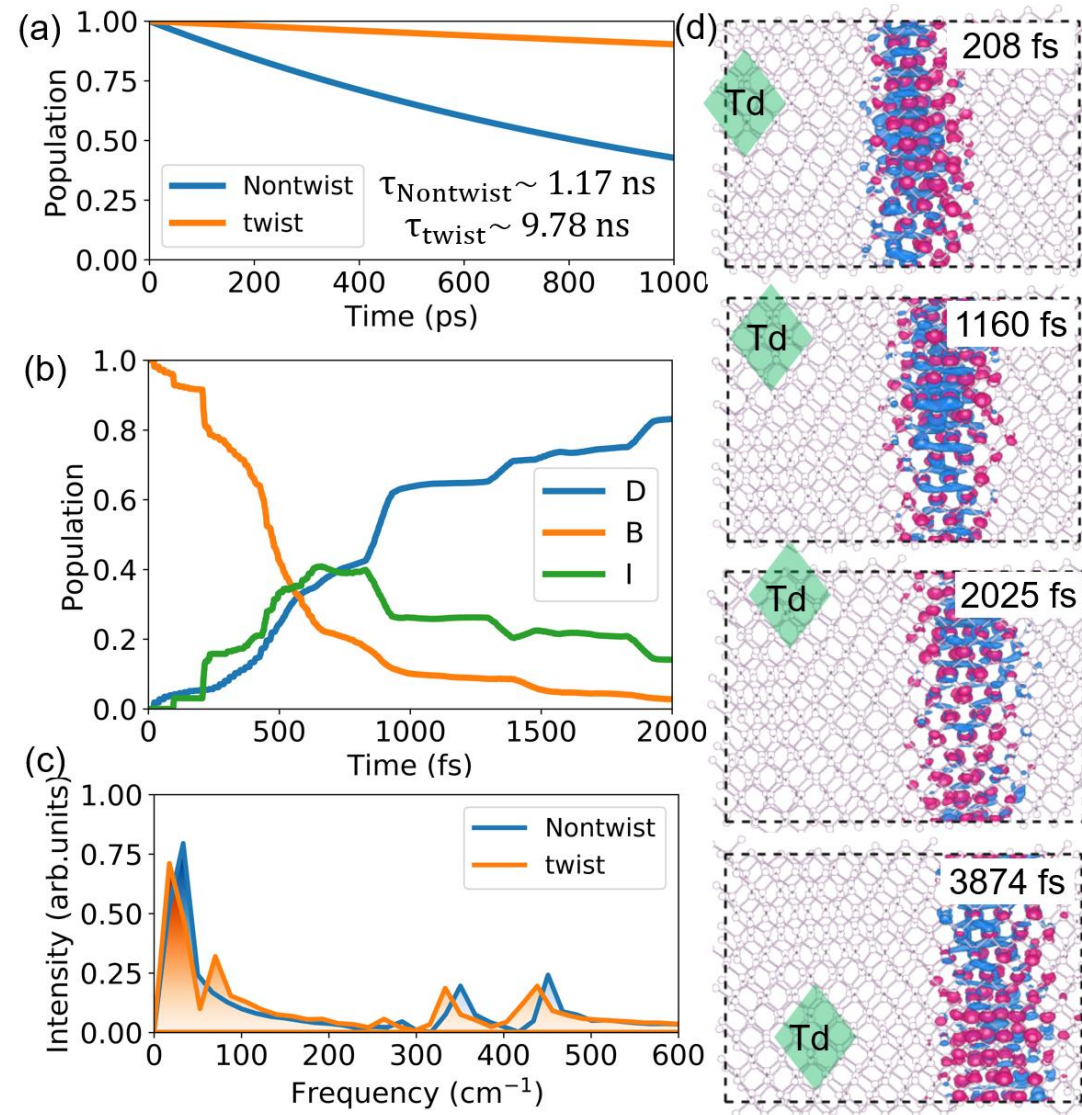
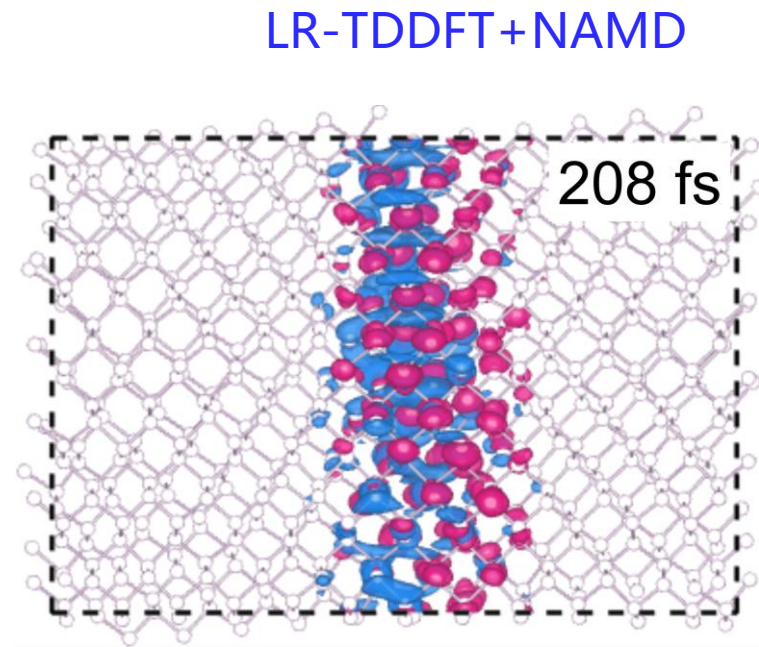
Pseudo-heterostructure and condensation of 1D moiré excitons in twisted phosphorene bilayers



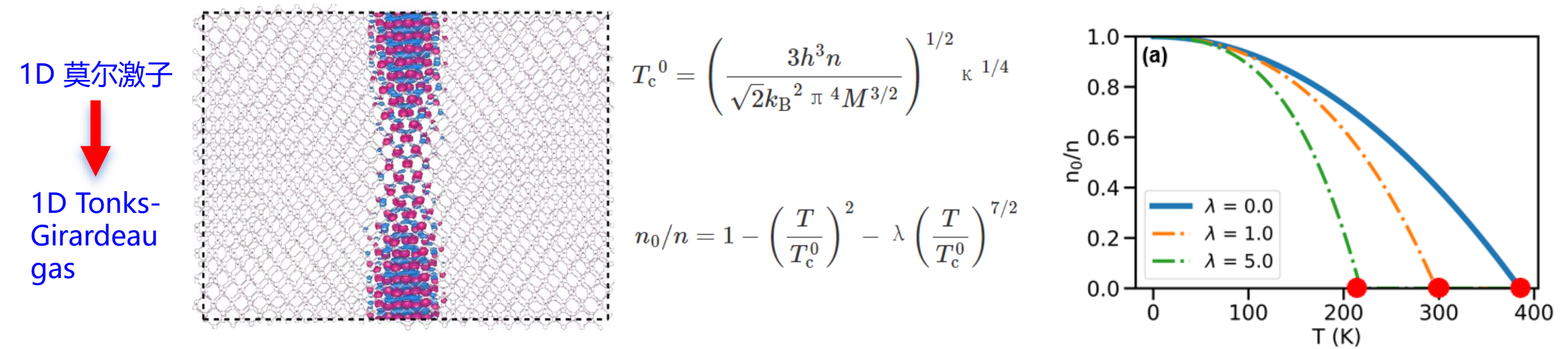
Symmetry forbidden dark moiré exciton in twisted bilayer BP



Pseudo-heterostructure and condensation of 1D moiré excitons in twisted phosphorene bilayers



Pseudo-heterostructure and condensation of 1D moiré excitons in twisted phosphorene bilayers



The 1D moiré excitons are predicted to exhibit Bose-Einstein condensation at high temperatures and may lead to exotic **Tonks-Girardeau Bose gases**

Summary

- We have developed a planewave pseudopotential LR-TDDFT method with screened and range-separated hybrid (SRSH) functionals capable to treat large periodic systems (more than 1000 atoms) with strong excitonic effect.
- We provided a first comprehensive study of moiré exciton properties from first-principles and explored how moiré exciton properties can be modulated by electric field, defect engineering and chemical tuning.
- We predicted that moiré excitons in Janus heterobilayers and tbP could be tuned for high-temperature BEC.

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